12th hacker series

Sexual reproduction in flowering plants

Flowering plants are known as angiosperm

Floriculture

- cultivation and marketing of flower

Panchanan Maheshwari:

Father of plant embryology





QUESTION

Intine layer is made up of

A. Cellulose and Pectin
B. Cellulose and chitin
C. Only with cellulose
D. Only with pectin

QUESTION

Part of exine where sporopollenin is not present are called

A. Germ pores
B. Sporopollen
C. Generative part
D. None of these

QUESTION

In 60 % angiosperm, pollen grains are shed at

A. 2-celled stage
B. 3-celled stage
C. 2-male gametes
D. Both (B) and (C)

QUESTION

The proximal end of filament is attached to

A. Thalamus
B. Stamen
C. Stigma
D. Anther

QUESTION

Which agent is most commonly used by grasses?

A. Wind
B. Water
C. Birds
D. ~~Biotic agent~~ Bees

QUESTION

Water pollination commonly occur in

A. Water hyacinth
B. Lily
C. *Vallisneria* and *Hydrilla*
D. Corn

QUESTION

Pollination occurring in closed flowers is:

A. Cleistogamy
B. Protogyny
C. Allogamy
D. Dicliny

QUESTION

Which of the following is known as common pansy?

A. *Commelina*
B. *Viola*
C. *Pisum*
D. *Vallisneria*

QUESTION

Chasmogamy refers to the condition where -

A. Flower remains closed
B. Flowers are absent
C. Flowers are open
D. Flowers are gamopetalous

QUESTION

Insect pollinated flowers usually posssess-

A. Brightly coloured pollen in large quantity
B. Dry pollens with smooth surface
C. Sticky pollen and rough surface stigma
D. Light coloured scented pollen lacking nectar

QUESTION

Which of the following animals act as pollinators?

A. Gecko lizard
B. Lemurs
C. Arboreal Rodents
D. ~~Both (A) and (C)~~ All are correct

QUESTION

What is the function of germ pore?

A. Emergence of radicle
B. Absorption of water for seed germination
C. Initiation of pollen tube
D. All the above

QUESTION

All the events from pollen deposition on the stigma until pollen tubes enter the ovules are together referred to as

A. Fertilisation
B. Double fertilisation
C. Pollen-pistil interaction
D. Pollination

Sexual reproduction

- Pre-fertilization events and structures
- Fertilization
- Post-fertilization events and structures

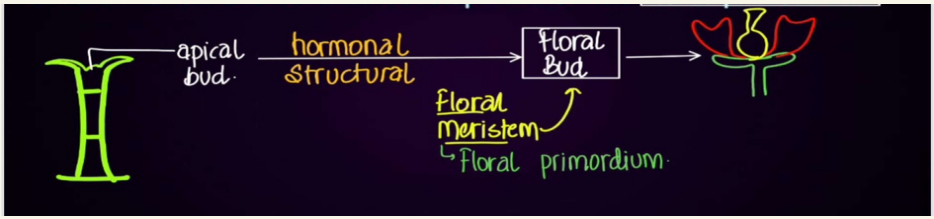
Flower

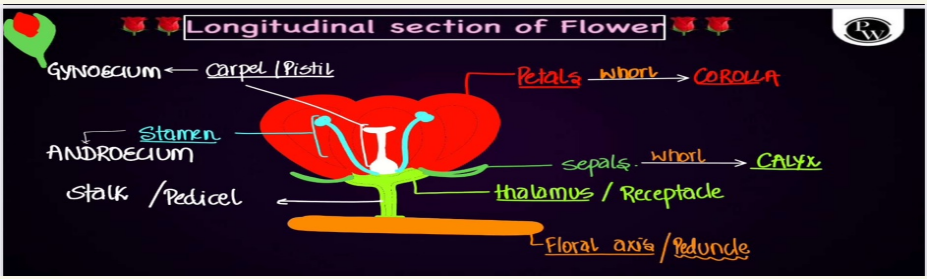
- a fascinating organ of angiosperms
- Flower is object of social, religious, cultural, ornamental, aesthetic

- for biologist, flowers are morphological and embryological, Marvel and site of sexual reproduction

Origin of flower

- several structural and hormonal changes occur, which lead to differentiation and further development of the floral primordium







Floral whorl

Essential whorl

→ Directly involved in reproduction

→ ANDROECIUM = ♂ rep. whorl

→ GYNOECIUM = ♀ rep. whorl

Non-essential whorl

→ Not directly involved in reproduction

CALYX

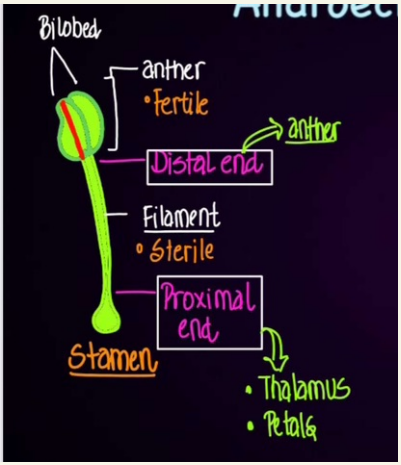
→ protect flower during bud stage

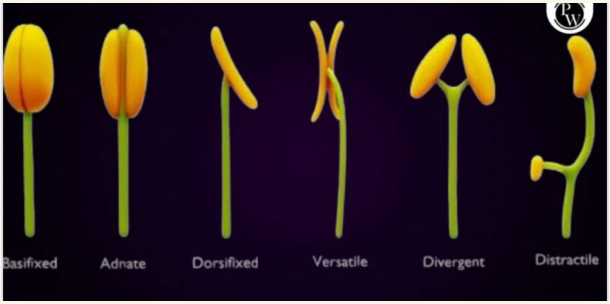
COROLLA

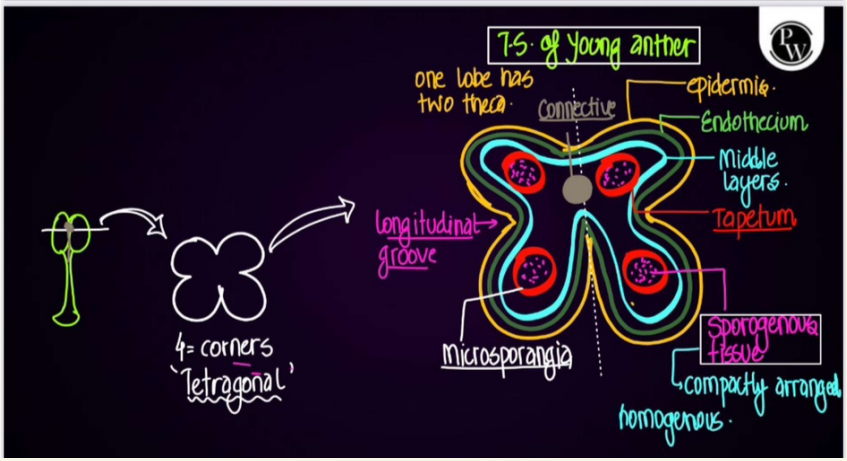
→ brightly coloured to attract pollinators

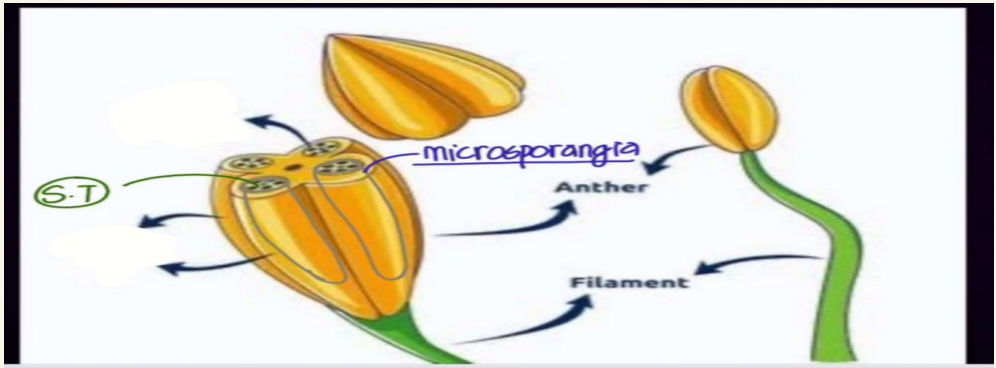
Androecium

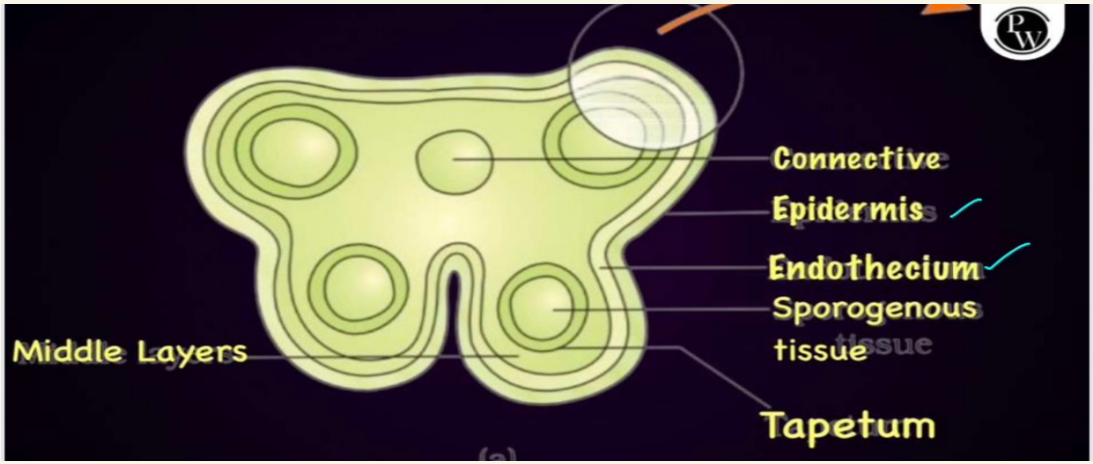
- whorl of stamens
- it represents the male reproductive organ
- Typical statement consist two parts
- Filament: long slender stalk
- anther: terminal generally bilobed structure
- Proximal end of anther is attached to thalamus or petal
- Number and length of stamen is variable from flower to flower
- Range in shape and attachment of anther in different flowers

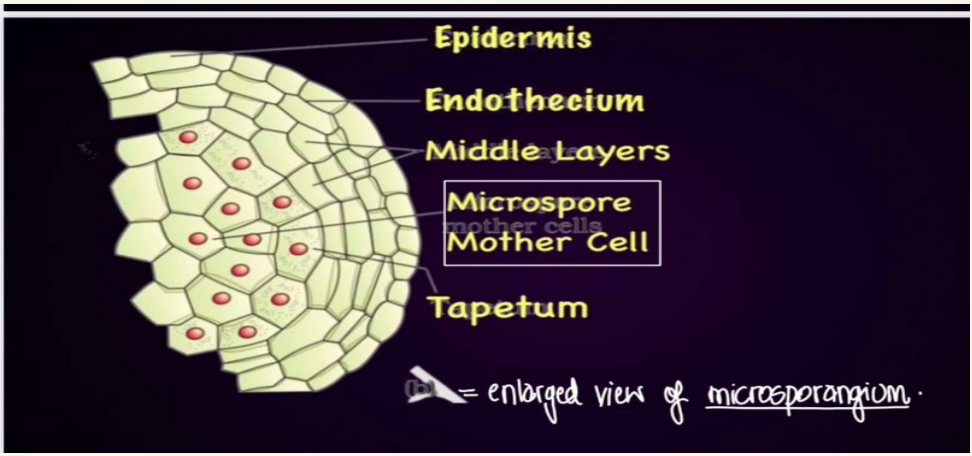


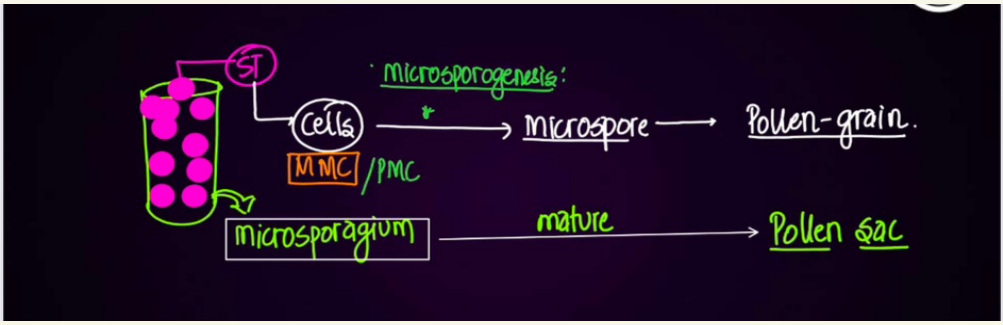


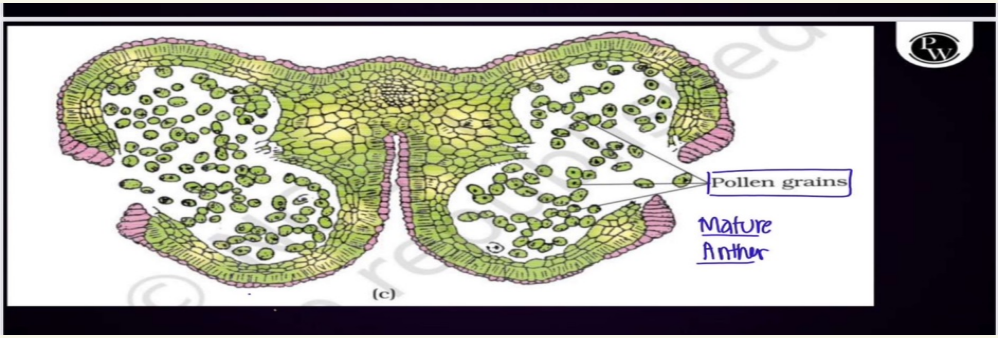



S.T
microsporangia
Anther
Filament


Connective
Epidermis
Endothecium
Sporogenous tissue
Middle Layers
Tapetum


Epidermis
Endothecium
Middle Layers
Microspore Mother Cell
Tapetum
= enlarged view of microsporangium.





Typical anther

- bilobed
- Dithecous (each lobe having two theca)
- tetrasporangiate (for microsporangia at each corner)
- tetragonal (having four corners)

First outer of three less → epidermis, endothecium, middle layer → protection & dehiscence of anther

Epidermis

- outermost layer of anther wall
- single layered
- Provide protection

Endothecium

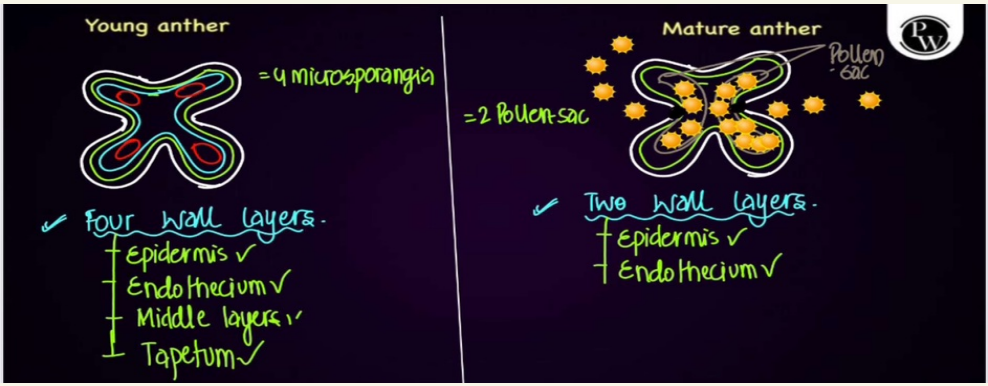
- it is present just below epidermis
- Single layered
- Main role in dehiscence of anther

Middle layer

- present between endothecium and Tapetum
- two three layers
- Consist paranchymatous cell

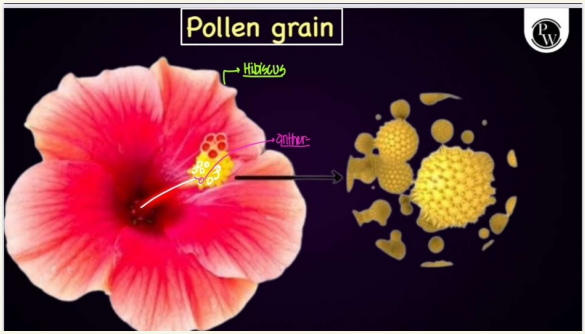
Tapetum

- most nutritive layer
- it nourish developing pollen grain
- Cells of this possesses dense cytoplasm and generally have more than one nucleus

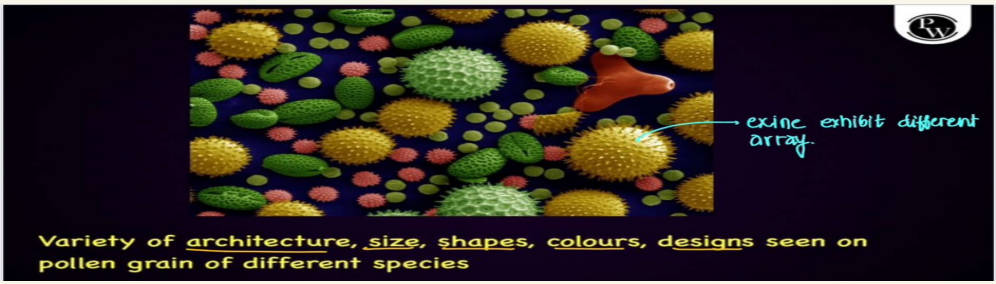


Micro sporogenesis

- the process of formation of Microspores from a pollen mother cell through meiosis is called micro sporogenesis
- The microspore as they are formed are arranged in cluster of four cells, the microspore tetrad
- as each cell of sporogenesis tissue is capable of giving rise to a microspore tetrad
- Each one is a potential pollen or microspore cell
- Has the anther, mature and dehydrate the microspores dissociate from each other and develop into Pollen grains

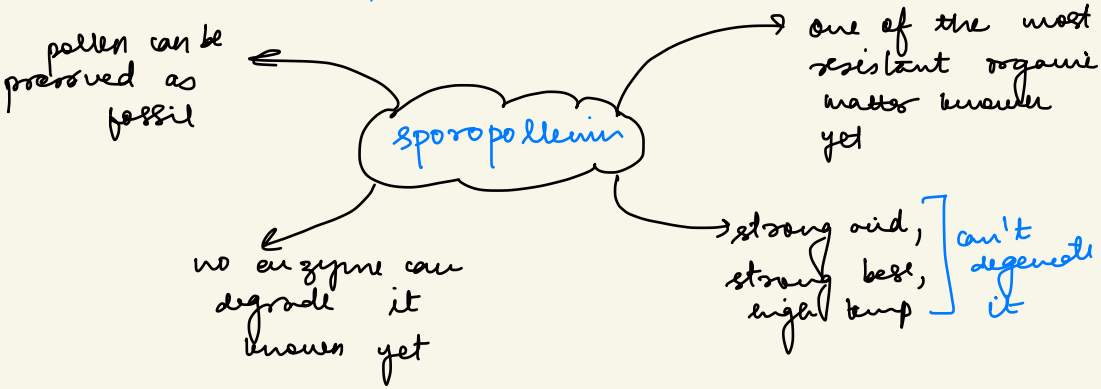






External development of pollen grain

- the pollen grains are generally spherical measuring about 25 to 50 micro metre in diameter
- it has a prominent two layered wall
- The hard outer layer is called the exine
- the inner wall of the pollen grain is called as intine

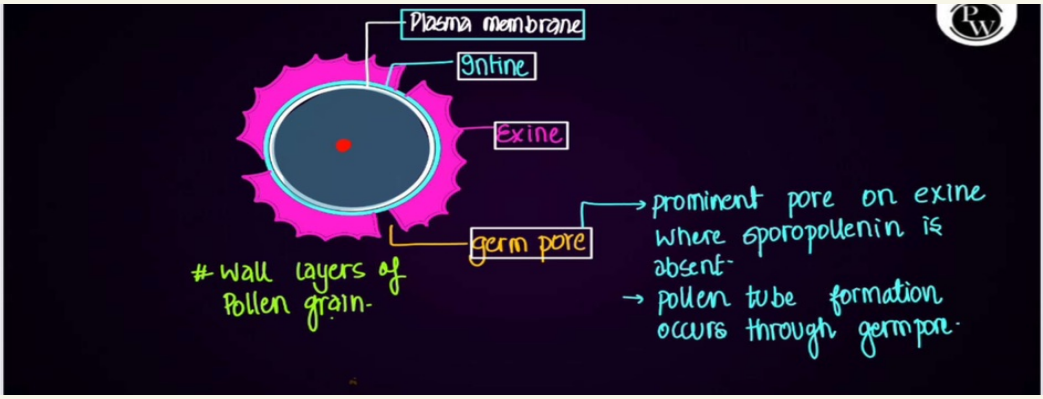


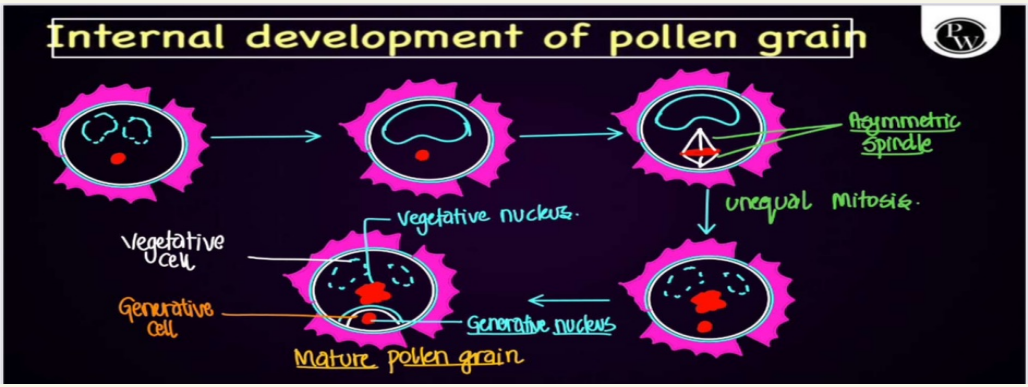
Exine

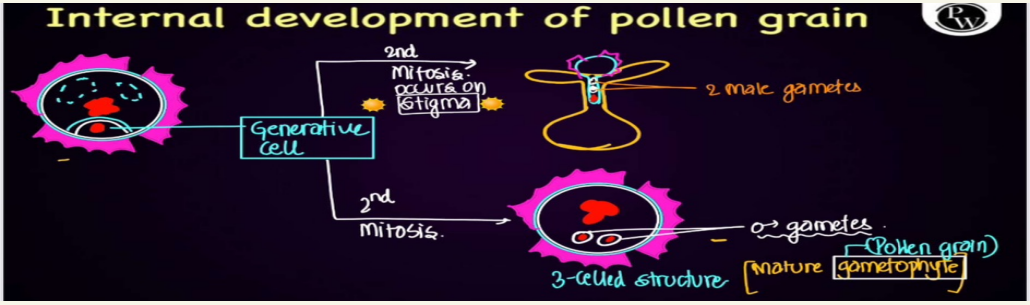
- outer layer
- Formed by tapetum
- Thick and discontinuous layer
- Made up of sporopollenin

Intine

- inner layer
- Formed by microspore
- Thin and continuous layer
- Made up of pectin and cellulose







→ 60% of angiosperm shed their pollen grain at 2-celled stage

→ 40% Of angiosperm shed their pollen grain at three cell stage

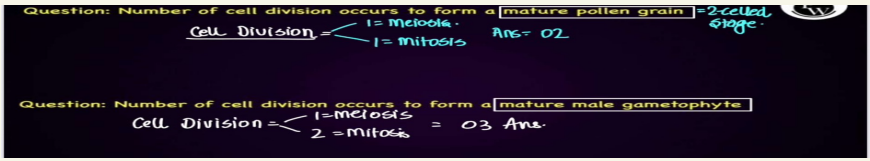
Microspore undergoes unequal mitosis

vegetative cell

- bigger cell
- Have abundant reserved. Food provide nourishment to generational cell
- single large irregularly shaped nucleus
- Helps in formation of pollen tube

Generative cell

- smaller cell
- Generate male gametes by mitosis
- Floats in cytoplasm of vegetative cell
- Single nucleus, dense cytoplasm
- Spindle shaped



Pollen allergy

- pollen grain that cause allergy is called as aero allergin
- Pollen grain of many plants cause severe
- Allergy
- bronchial affliction
- hayfever
- Rhinitis
- Chronic respiratory disorder(asthma)

Pollen products

- Pollen grains are rich in nutrients
- Pollen grains are used as food supplements in western countries

- used by athletes and race horses to improve performance
- Pollen grains are used to make body lotion and sunscreen



Pollen viability

- the capability of Pollen to germinate on stigma and fertilise
- Highly variable
- Factors affecting viability → Internal factors = genetic composition
  → external factors = prevailing temp. humidity

- eg) wheat, rice = viability (30 min)

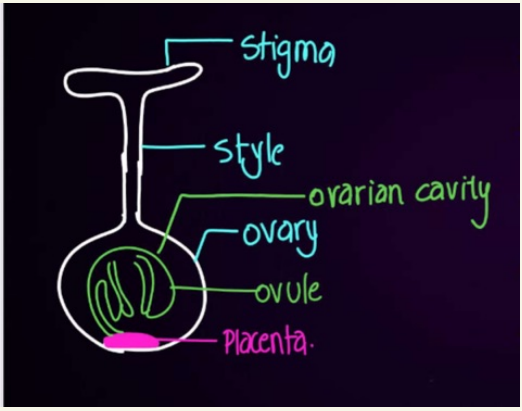
S – Solanaceae
R – Rosaceae
L – Leguminosae
] viability = months

Pollen banks

- storage of Pollen grains for years
- Reserved in liquid nitrogen at -196°C called as cryopreservation
- Used in crop breeding programmes

Gynoecium

- stigma serves as a landing platform for pollen grains
- The style is the elongated slender part beneath the stigma
- The basal bulged part of the pistil is the ovary
- Inside the ovary is the ovarian cavity (locule)
- The placenta is located inside the ovarian cavity



Number of carpels

Monocarpellary
consists of single carpel
eg- mango
pea
coconut

Multi | polycarpelly
consists many carpel

Apocarpous
free carpel
→ Rose
→ Lotus
→ Michelia

Syncarpous
fused carpel
→ Hibiscus
→ papaver somniferous



Number of Ovules inside a carpel

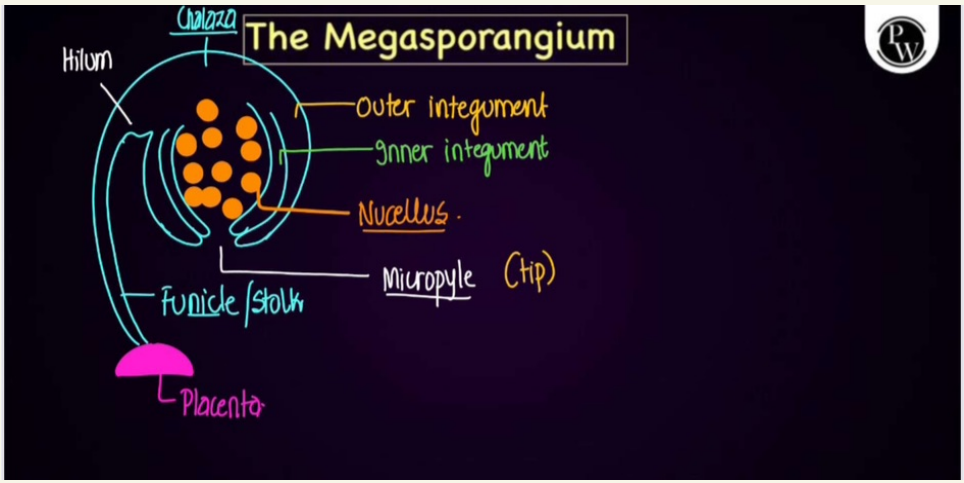
One

- mango
- wheat
- paddy (Rice)

Many

- watermelon
- papaya
- guava
- orchids
- Ficus
- strigea
- orobanche





FUNICLE

The ovule is a small structure attached to the Placenta by means of a stalk called funicle.

HILUM

The body of the ovule fuses with Funicle in the region called hilum.

- represents the junction between ovule and funicle

INTEGUMENT

Each ovule has one or two protective envelopes called integuments.

unitegmic / Bitegmic

MICROPYLE

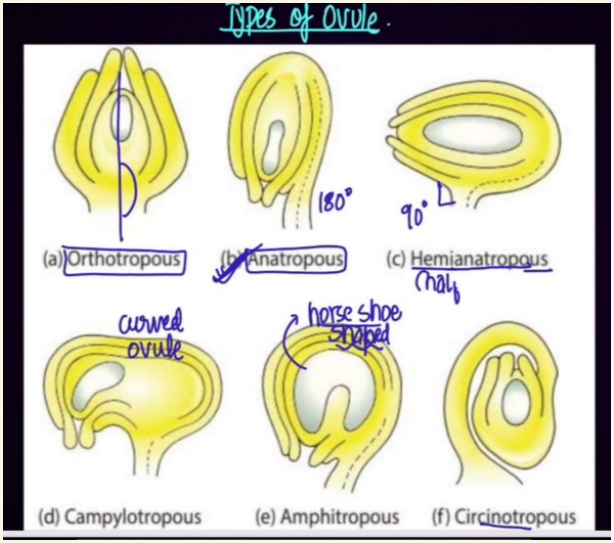
Integuments encircle the nucellus except at the tip where a small opening called the micropyle

CHALAZA

Opposite the micropylar end, is the chalaza, representing the basal part of the ovule.

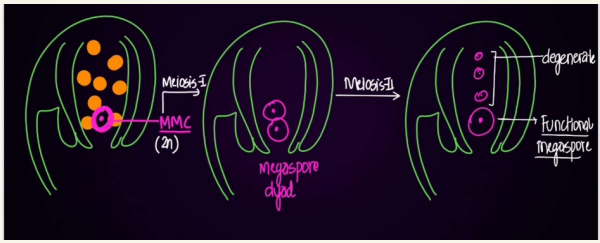
NUCELLUS

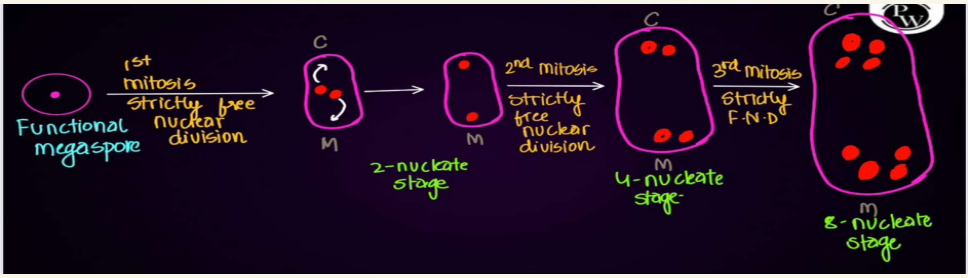
Enclosed within the integuments is a mass of cells called the nucellus. Cells of the nucellus have abundant reserve food materials

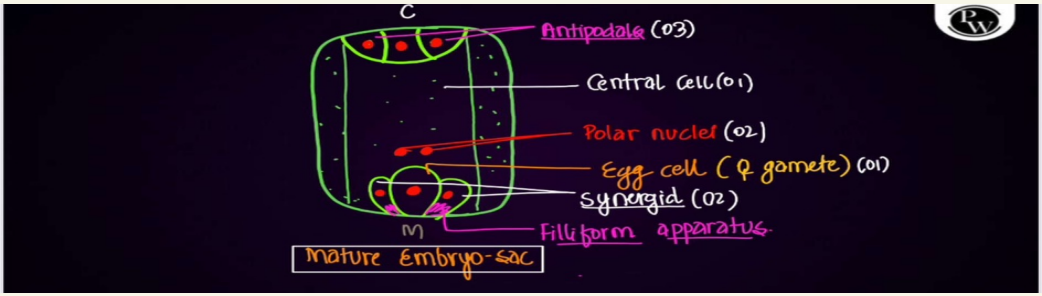


Megasporogenesis

- process of formation of megaspore from megaspore mother cell
- Three megaspore degenerate, one megaspore remain functional
- One functional megaspore further develops in female gametophyte that is embryo sac . Such development is called monosporic development







Female gametophyte

- the nucleus of a functional megaspore divides mitotically to form two nuclei, which move to the opposite poles, forming the two nucleate embryo sac
- Two more sequential mitosis nuclear division result in the formation of the four nucleate and later the eight nucleate stages of the embryo sac
- mitotic division are strictly free nuclear that is nuclear division are not followed immediately by cell wall formation
- After the eighth nuclear stage, cell walls are laid down leading to the organisation of the typical female gametophyte or embryos sac
- Six of the eight nuclei are surrounded by the cell walls and organised into cells
- The remaining two nuclei called polar nuclei are situated below the egg apparatus in the large Central cell
- There is a characteristic distribution of the cells within the embryo sac

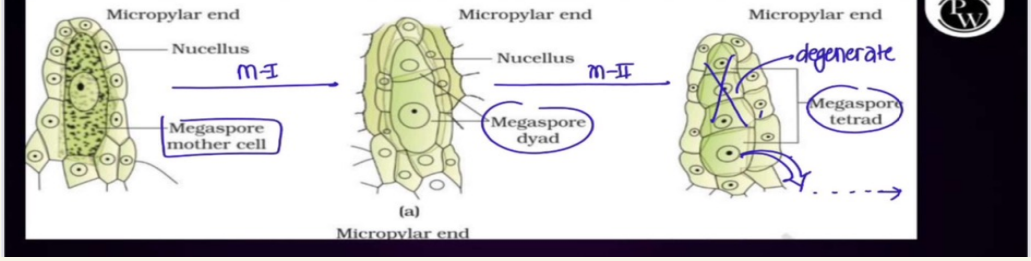
1. Three cells are grouped together at the micropylar end and constitute the egg apparatus.

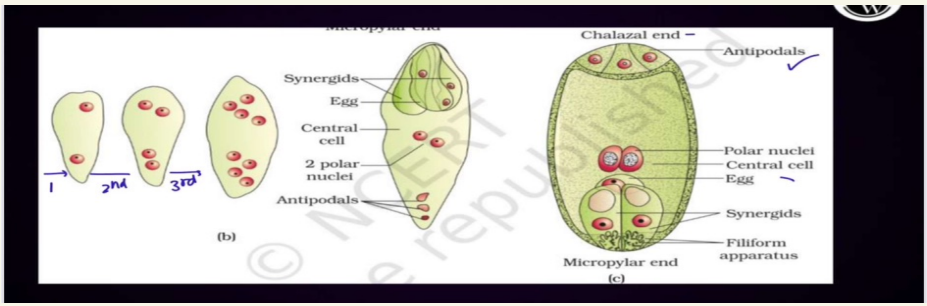
2. The egg apparatus, in turn, consists of 02 synergids and 01 egg cell.

3. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid.

4. Three cells are at the chalazal end and are called the antipodals.

5. Thus, a typical angiosperm embryo sac, at maturity, though 08-nucleate is 07-celled.

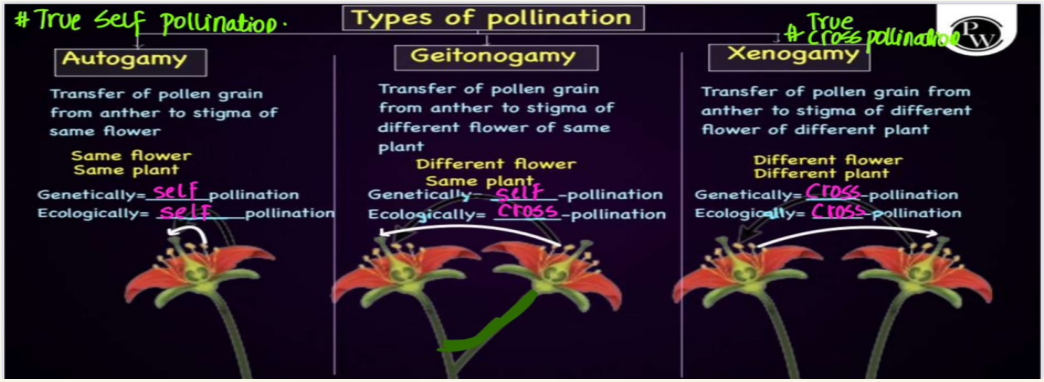


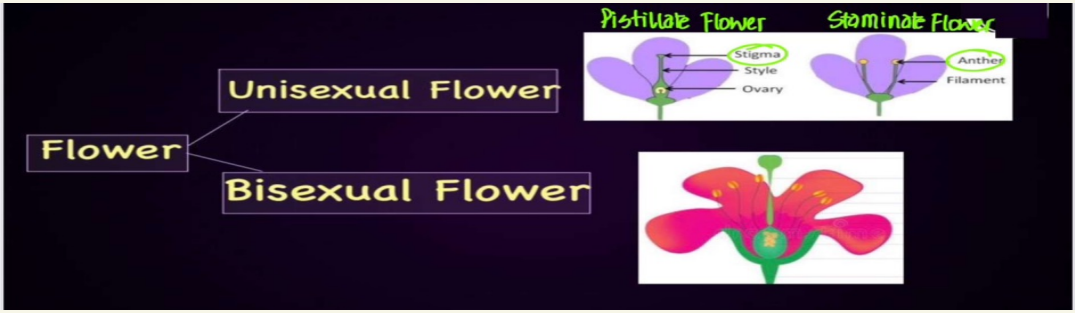


Pollination

- transfer of pollen grain from anther to stigma







MONOECIOUS

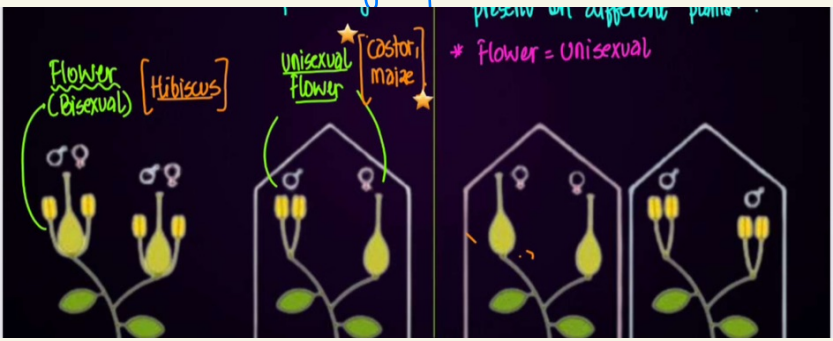
single house

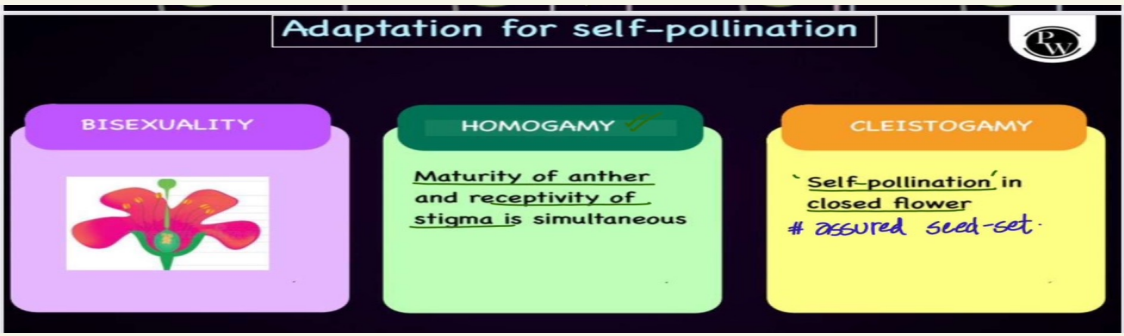
→ single plant → Both → ♂ sex organ
→ ♀ sex organ

DIOECIOUS

two house

→ ♂ & ♀ sex organ in different plants.







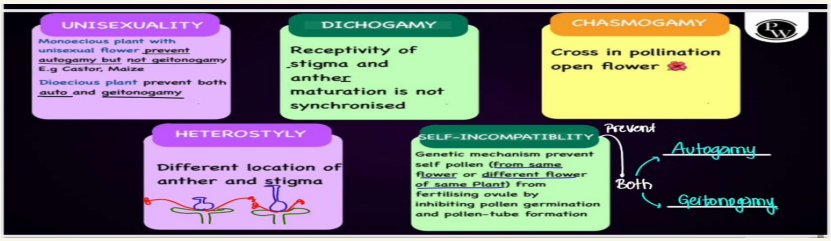
majority of flowers = hermaphrodite | Bisexual

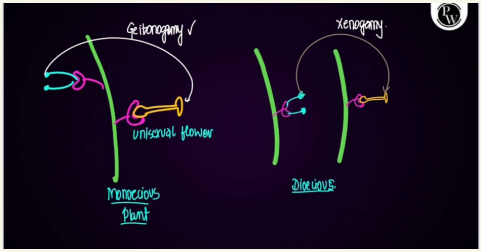
(continued) more → self pollination

↓

Inbreeding depression

Adaptation for cross pollination/ outbreeding devices/ convance for cross pollination





Agents of pollination

- ABIOTIC
  - Wind — Anemophily
  - Water — Hydrophily
- BIOTIC
  - Animal — Zoophily
    - Insect — Entomophily
    - Bird — Ornitho-phily

ANIMAL >> Wind >> Water

↓ mfc

Insect — rare

Bee is the most common among insects.

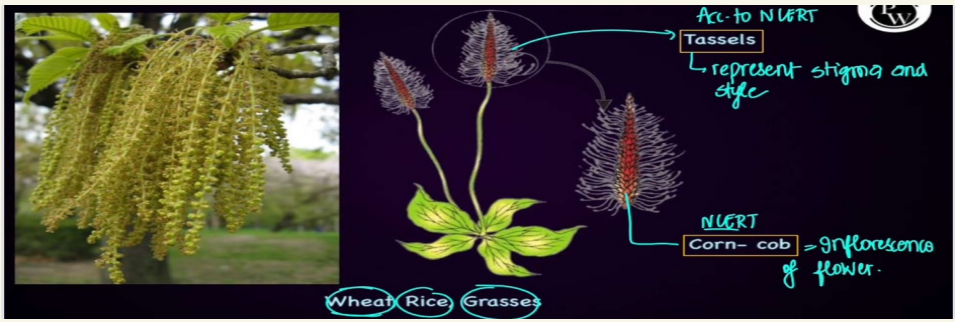
→ Fav. colour = yellow

pollination in yellow flower

Most common among abiotic agent is wind

Anemophily

- waterless
- Colourless
- odourless
- usually contain 1 ovule
- Flowers are grouped in inflorescence
- long, sticky, feathery stigma
- light weight, non-sticky
- dry formed in large amount



Not all hydrophytes show hydrophily.
eg- water lily, lotus

- colourless
- odourless
- long stigma
- only 30 genera (mostly monocots)
- most rare type

Hydrophily

→ pollen grain is covered by mucilaginous coat to prevent it from wetting
→ long & ribbon like



Hydrophily

Hypohydrphily

- pollination occurs below water
- Example- Hydrilla, zostera

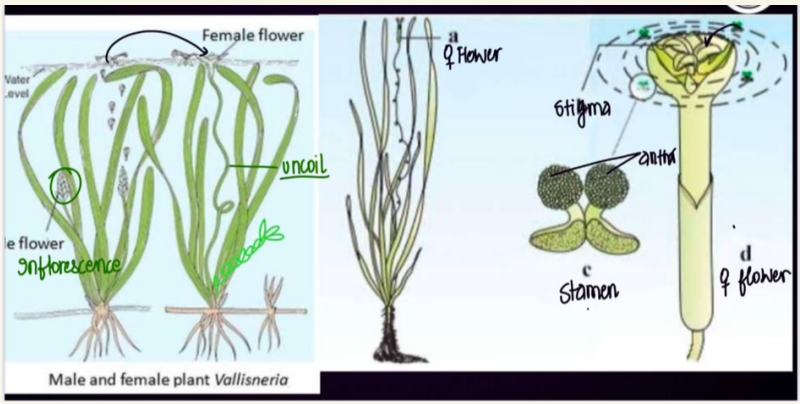
Epihydrophily

- pollination occurs on surface of water
- Example - vallisneria

- sub-merged
- fresh H_2O

female flower reaches the surface

male flower are pollen grain are released on the surface

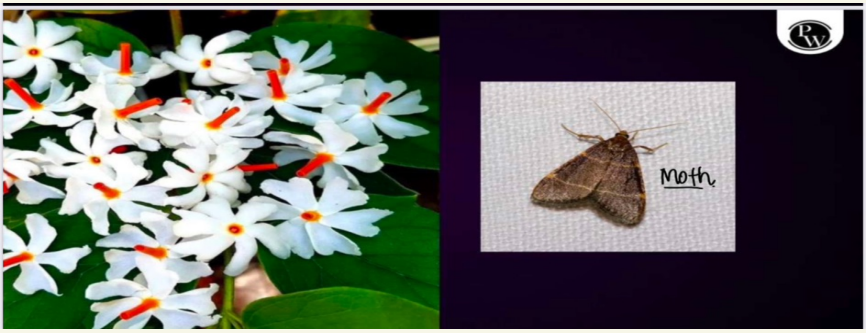


Male and female plant *Vallisneria*

Zoophily

Entomophilous flower

- provide floral rewards to insects: nectar, pollen grain, and Oviposition site
- Plants which show flowering at night, highly scented, usually white and coloured nectar is present pollinated by moth
- Colourful with odour fragrance
- Foul odour beetle flies
- Large, if small, make influorescence to make them conspicuous
- Stigma is sticky or non-sticky
- Pollen grain is sticky due to presence of pollen kitt

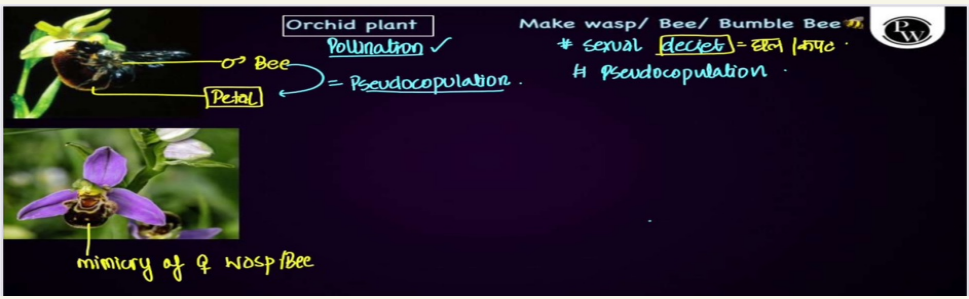


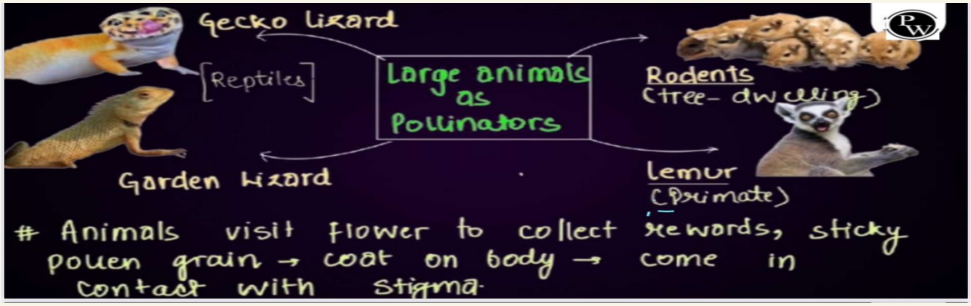


Pollen nectar robbers

- insects, receive floral rewards, but do not help in pollination





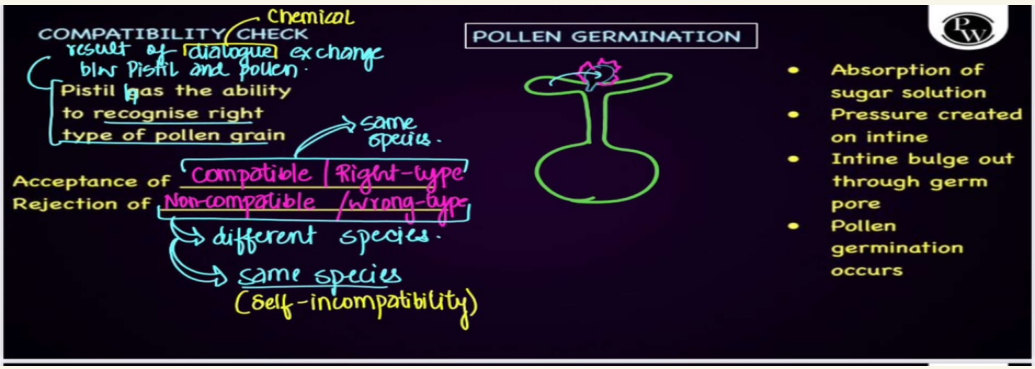


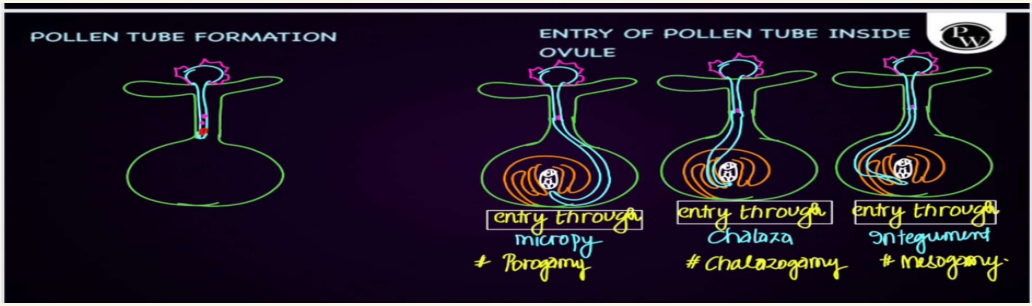
Pollen pistil pollination

- from Pollen deposition on stigma to entry of pollen tube inside ovule
- Dynamic process involving Pollen recognition, followed by promotion or inhibition of pollen grain

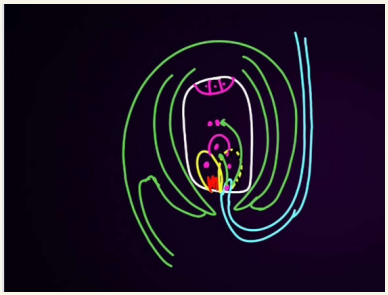
steps

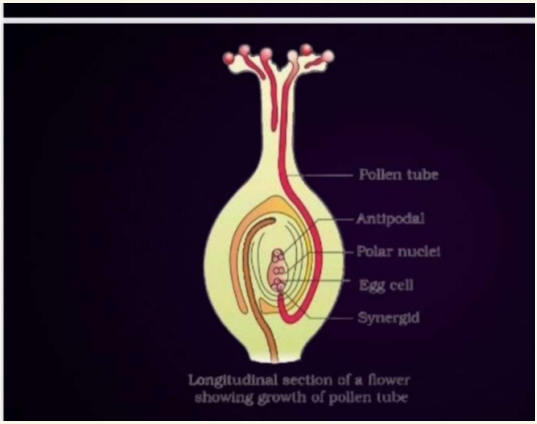
- compatibility check
- pollen germination
- pollen tube formation
- entry of pollen tube inside ovule

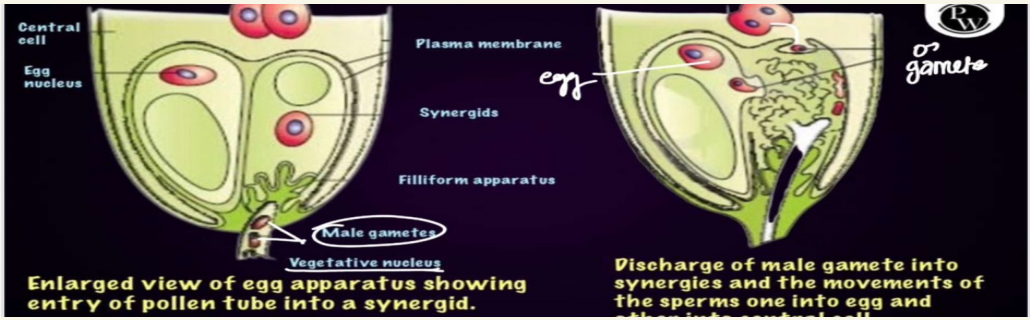










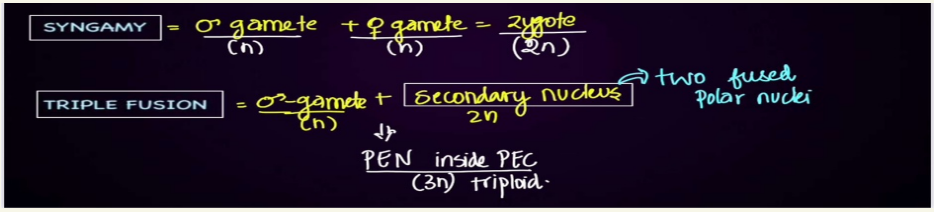


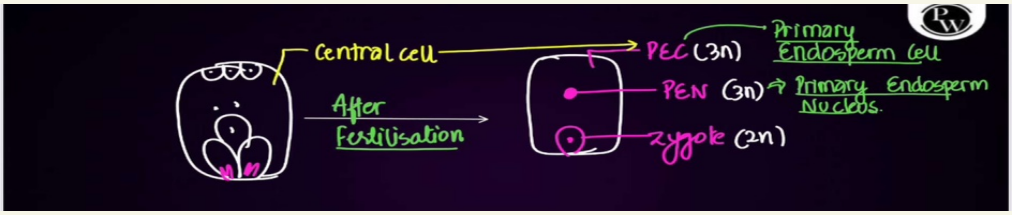
Entry of Pollen tube inside of ovule occurs through- micropyle mostly

Entry of Pollen tube inside embryo sac occurs through- synergids

Double fertilisation

- characteristic feature of angiosperms
- Fertilisation occur, occur twice





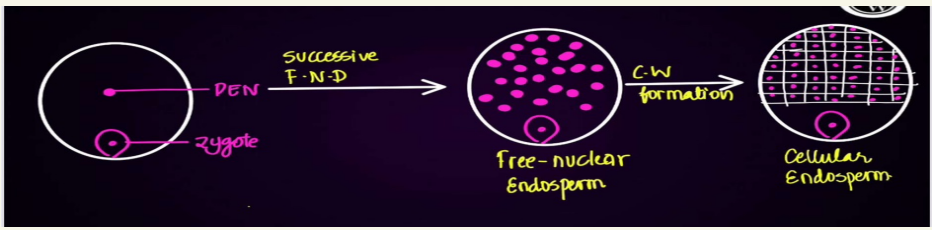
polar nuclei = n + n

↓ fusion

secondary nucleus = 2n

Endosperm

- product of triple fusion
- Cells are triploid
- Filled with reserve food material and are used for the nutrition of developing embryo
- PEN undergo successive free nuclear division and give rise to free nuclear endosperm
- Later, cell wall formation occurs and form cellular endosperm

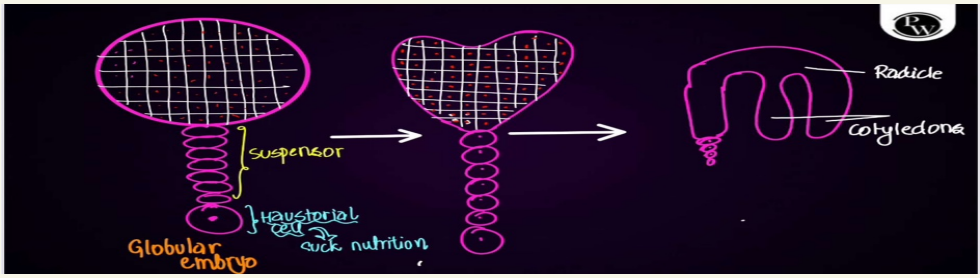


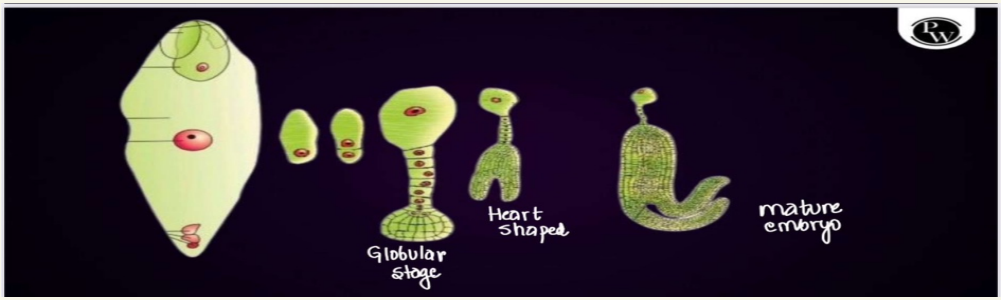


Embryonic development

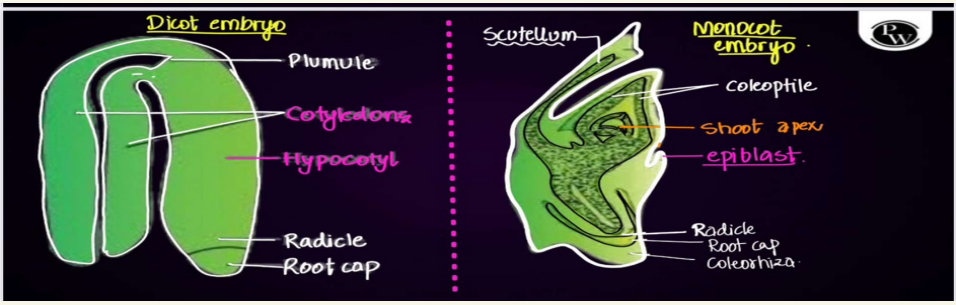
- endosperm development proceeds, embryonic development







- embryo develops at micropyle of embryo, where zygote is located
- zygote Divides only after certain amount of endosperm is formed
- This is an adaptation to provide assured nutrition to developing embryo
- Early stages are same in both monocots and dicots

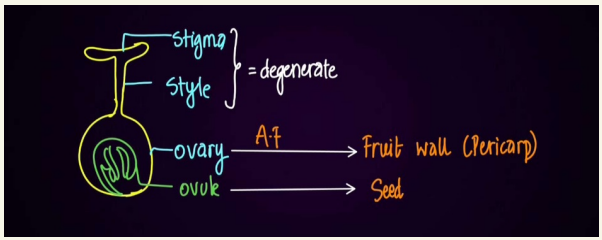


Dicot embryo

- consist embryonal axis
- Two cotyledons
- Portion of embryonal axis above Cotyledons is called epicotyl, which is terminated by plumule
- Root tip is covered with root cap

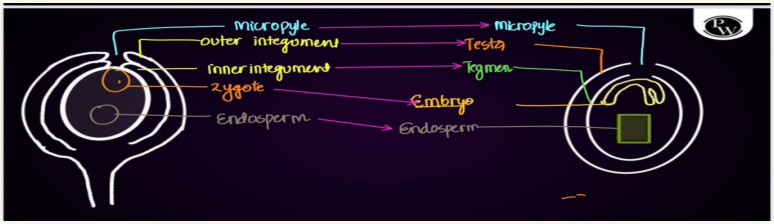
Monocot embryo

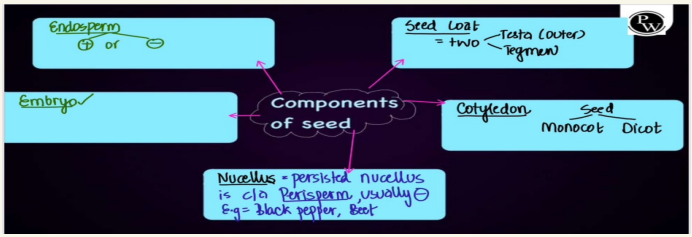
- possess one cotyledons
- in grass, family, single cotyledons is called as scutelum
- Radical and root cap are enclosed within an differentiated sheet called as coleorhiza
- Shoot apex has few leaf Primordia enclosed with hollow foliar structure called as coleoptile



Seed

- final product of sexual reproduction
- Seed is fertilised ovule





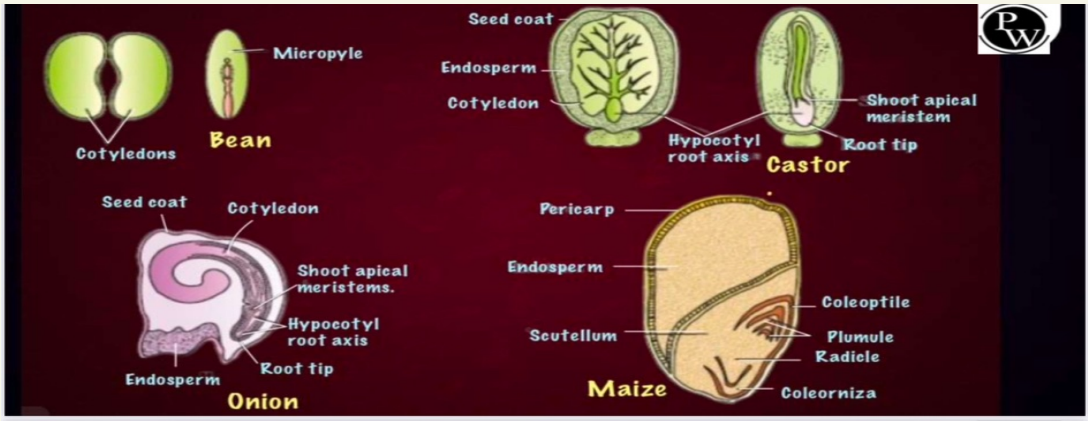
Types of seed

Endosperm seed

- albuminous seed
- endo form is persisted in mature seed
- Usually present in monocots
- Example- wheat rice maize barley

Non-endosperm seed

- ex albuminous seed
- Endosperm is not present
- Usually present in dicots
- example- Pea bean gram



As embryo matures

↓

water content ↓

↓

seed becomes relatively dry (10-15% H_2O by mass)

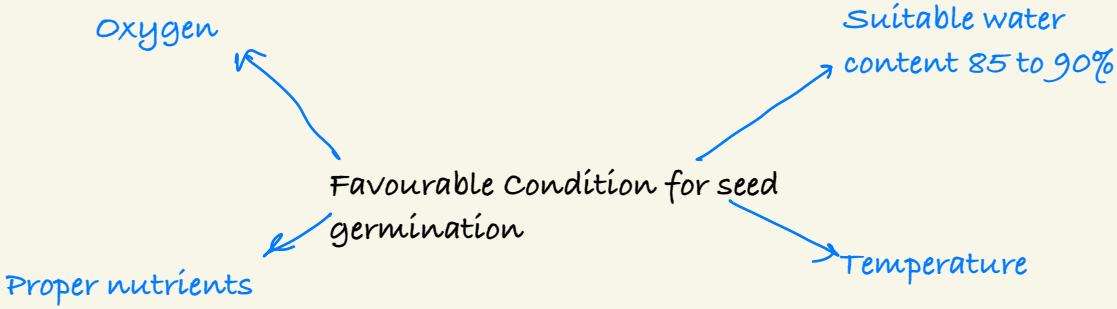
↓

metabolic reaction slows down

↓

embryo enters state of inactivity

Dehydration and dormancy are crucial for storage of seed

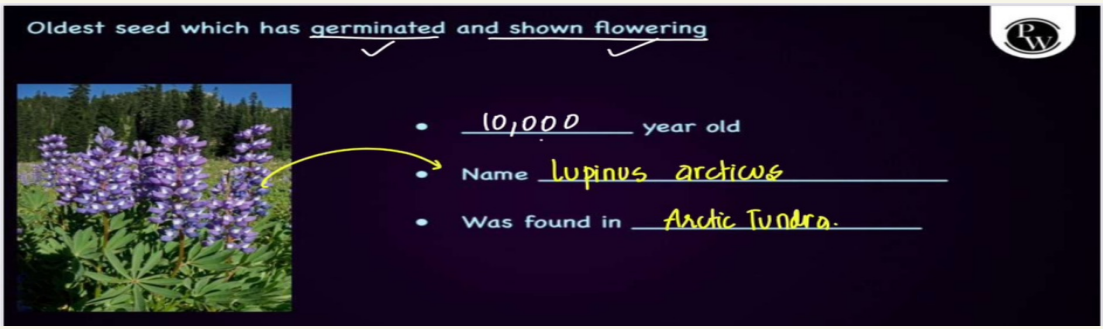


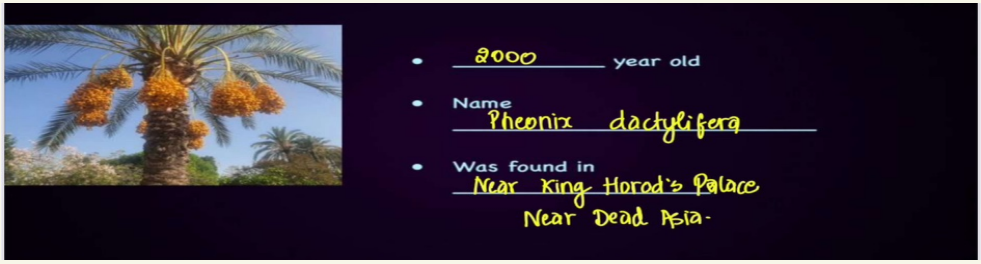
Advantages of seed formation for plants

- fertilisation and pollination is independent of water. Seed formation is more dependable.
- Seed has hard seed coat, which protect embryo
- Provide nutrients to young seeds until it starts photosynthesis
- Seed has better seat dispersal strategies
- Being product of sexual reproduction seed generate new genetic combinations, leading to variation does lead to evolution

Seed viability

- seed remain capable of germination
- highly variable

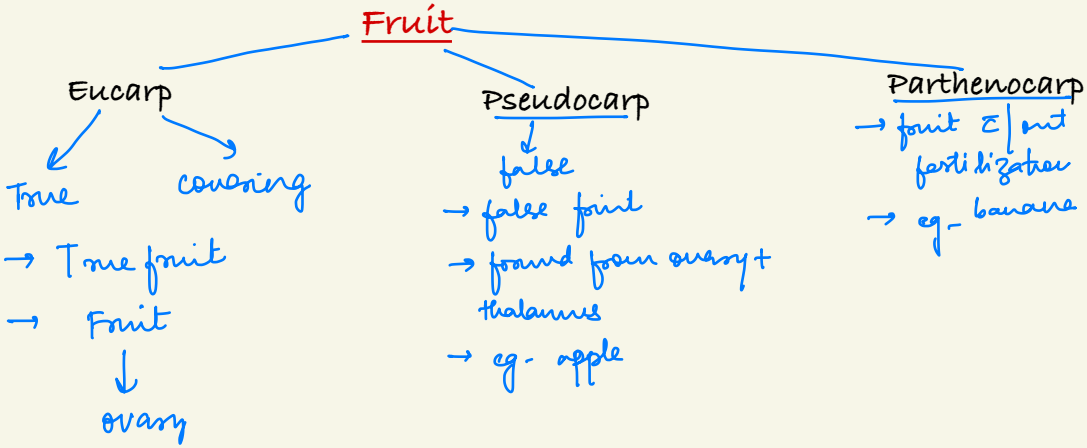


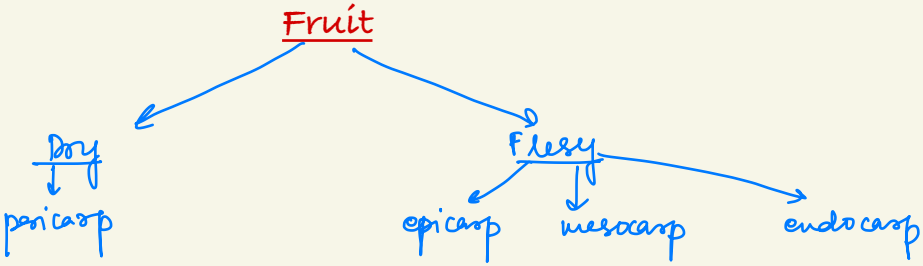


Fruits with tiny seeds

- orchids
- Ficus
- orobanche
- Strigo

] parasite



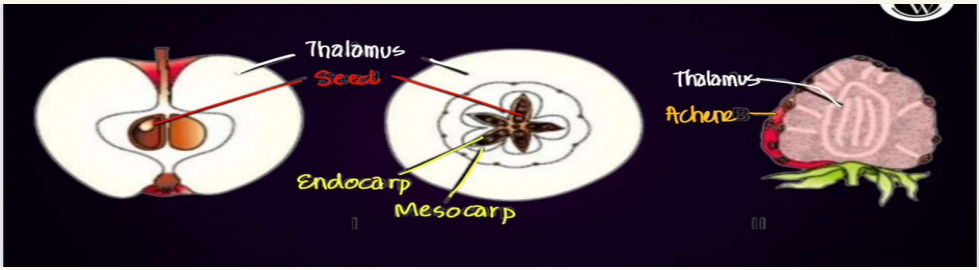


Drupe

- mono carpellary superior ovary
- Mesocarp- fleshy/fibrous

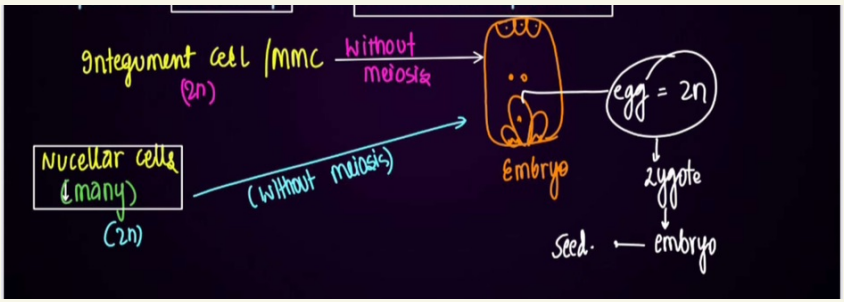
- epicarp- thin
- Endocarp - Stony





Apomixis

- formation of seed without fertilisation
- it is a sexual reproduction, which mimic sexual reproduction

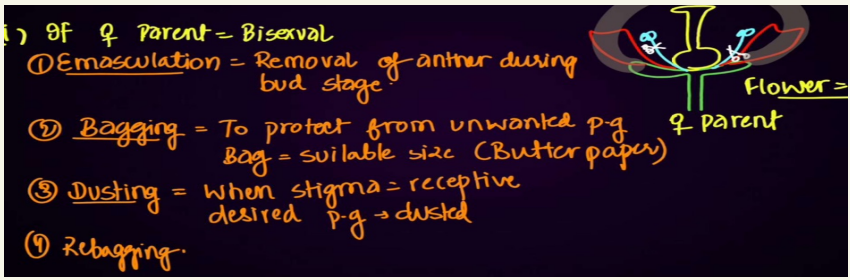


- in some species diploid Egg cell is formed without reductional division and develops into embryo without fertilisation
- in citrus and mango, some nucellar cells surrounding embryos starts dividing protrude into embryo sac and develop into embryos inside single ovule
- Occurrence of more than one embryo in a seed is referred to as polyembryony

Apomictic seed

- it plays a important role in hybrid seed production
- The method of producing hybrid seeds by cultivation is very expensive for farmers
- Also by showing hybrid seats, it is difficult to maintain hybrid characters as characters segregated during meiosis
- apomixis revent the loss of specific characters in the hybrid
- Also it is a cost-effective method for producing seeds

Artificial hybridisation



To produce 1600 seeds, the number of meiotic divisions required will be

A. 2400
B. 2000
C. 1600
D. 1800

$n + \frac{n}{4}$ $\quad 1600 + \frac{1600}{4}$

n = no. of seed

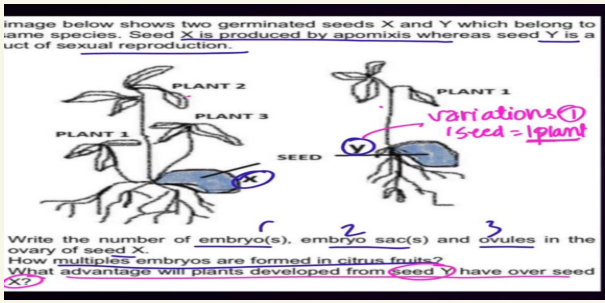


image below shows two germinated seeds X and Y which belong to ame species. Seed X is produced by apomixis whereas seed Y is a uct of sexual reproduction.

Write the number of embryo(s), embryo sac(s) and ovules in the ovary of seed X.
How multiples embryos are formed in citrus fruits?
What advantage will plants developed from seed Y have over seed X?

1 2 3

variations ①
1 seed = 1 plant

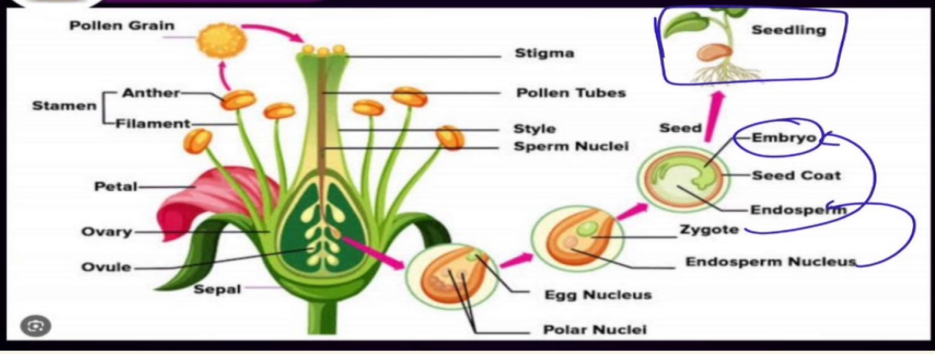
x = 1 = 03
2 = 01
3 = 01

Competency Based Qu

(i) Explain the significance of each of the following features present in plants given below: (Outbreeding devices)
a) In rose-bay plant the stamens ripen before the stigma.
b) In certain species of primrose, the flowers have short stamen and long style.
c) The bisexual flower of mustard exhibits rejection of self-pollen grain.
(ii) Explain how autogamy is prevented in castor and papaya plant respectively.

Competency Base

Summary


Pollen Grain
Stamen
Anther
Filament
Petal
Ovary
Ovule
Sepal
Stigma
Pollen Tubes
Style
Sperm Nuclei
Egg Nucleus
Polar Nuclei
Zygote
Endosperm Nucleus
Seed
Embryo
Seed Coat
Endosperm
Seedling